

Discrete Homework Seven

1. We have the following:

$$x \equiv 2 \pmod{5}$$

$$x \equiv 3 \pmod{7}$$

$$x \equiv 4 \pmod{8}$$

Through $M = (m_1)(m_2)(m_3)$, we find:

$$M = (5)(7)(8) = 280$$

Using the formula $M_n = M/m_n$, we know:

$$M_1 = 280/5 = 56$$

$$M_2 = 280/7 = 40$$

$$M_3 = 280/8 = 35$$

Knowing $(M_n)(M_n^{-1}) = 1 \pmod{m_n}$, we find the following:

$$56(M_1^{-1}) = 1 \pmod{5}$$

$$M_1^{-1} = 1$$

$$40(M_2^{-1}) = 1 \pmod{7}$$

$$M_2^{-1} = 3$$

$$35(M_3^{-1}) = 1 \pmod{8}$$

$$M_3^{-1} = 3$$

Using the formula $x = (a_1M_1M_1^{-1} + a_2M_2M_2^{-1} + a_3M_3M_3^{-1}) \pmod{M}$, we find:

$$x = (2(56)(1) + 3(40)(3) + 4(35)(3)) \pmod{280}$$

$$x = (892) \pmod{280}$$

$$x = 52$$

Therefore, we arrive at our answer of $x = 52$.

2. The bolded text decrypted is:

“pineapple pizza is simply superior than normal pizza”

3. The statement encrypted into a shift cipher is:

“IFEEZV DRUV KYZJ HLVJKZFE”

4. The statement encrypted is:

SIAOLEMATDOHINSXEXXX

5. The statement decrypted is:

twentyfiftyx

HOWEVER, we know an extra x is added to the end to ensure the blocks are of size three, so the decrypted text is “twentyfifty”

6. Plaintext: “MATHISFUN”

Key: (77, 17)

Note: $25 < 77 < 2525$, therefore, block size is 2 digits

String of digits: 120019070818052013

Blocky digits: 12 00 19 07 08 18 05 20 13

$$12^{17} \bmod 77 = 45$$

$$00^{17} \bmod 77 = 00$$

$$19^{17} \bmod 77 = 24$$

$$07^{17} \bmod 77 = 28$$

$$08^{17} \bmod 77 = 57$$

$$18^{17} \bmod 77 = 72$$

$$05^{17} \bmod 77 = 03$$

$$20^{17} \bmod 77 = 48$$

$$13^{17} \bmod 77 = 62$$

Therefore, we have the ciphertext of 450024285772034862.

7. We have the ciphertext of “09 11 49 52 49 33 52 02 00 10 17 39 08 13 09 52 02 17 52 10” with the encryption key of (55, 23). We will start by finding prime numbers p and q such that $N = 55$. $55 = (p)(q) = (5)(11)$. Therefore, $p = 11$ and $q = 5$. We know $(p - 1)(q - 1) = (11 - 1)(5 - 1) = (10)(4) = 40$. Therefore, $d(23) \equiv 1 \pmod{40}$. Hence, $d = 7$. Proof? Trivial. Now, we have the form $m = c^7 \pmod{55}$.

$$09^7 \pmod{55} = 4 \text{ E}$$

$$11^7 \pmod{55} = 11 \text{ L}$$

$$49^7 \pmod{55} = 14 \text{ O}$$

$$52^7 \pmod{55} = 13 \text{ N}$$

$$29^7 \pmod{55} = 14 \text{ O}$$

$$33^7 \pmod{55} = 22 \text{ W}$$

$$52^7 \pmod{55} = 13 \text{ N}$$

$$02^7 \pmod{55} = 18 \text{ S}$$

$$00^7 \pmod{55} = 00 \text{ A}$$

$$10^7 \pmod{55} = 10 \text{ K}$$

$$17^7 \pmod{55} = 8 \text{ I}$$

$$39^7 \pmod{55} = 19 \text{ T}$$

$$08^7 \pmod{55} = 2 \text{ C}$$

$$13^7 \pmod{55} = 7 \text{ H}$$

$$09^7 \pmod{55} = 4 \text{ E}$$

$$52^7 \pmod{55} = 13 \text{ N}$$

$$02^7 \pmod{55} = 18 \text{ S}$$

$$17^7 \pmod{55} = 8 \text{ I}$$

$$52^7 \pmod{55} = 13 \text{ N}$$

$$10^7 \pmod{55} = 10 \text{ K}$$

PLEASE NOTE: IF THE ABOVE MODULUS ANSWER WAS GREATER THAN 25, THEN THE FOLLOWING FORMULA WAS USED:

$$X \text{ MOD } 25 = n$$

Therefore, we arrive at the decrypted message of “ELONOWNSAKITCHENSINK”

8. Use mathematical induction, prove that $3 + 7 + 11 + \dots + (4n-1) = n(2n + 1)$ for all positive integers n .

To start, we must establish the base case for when $n = 1$, as 1 is the first positive integer. Notice that the Left-hand side becomes 3 as $4(1) - 1 = 3$ and the Right hand side becomes 3 as well since $1(2(1) + 1) = 3$. Since the left hand side equals the right hand side the base case has been proven. Now we assume the inductive hypothesis for some positive integer k such that $3 + 7 + 11 + \dots + (4k-1) = k(2k + 1)$. We seek to show that if the statement is true for k , then the statement is true for $k + 1$. Thus, we want to show that $3 + 7 + 11 + \dots + (4k-1) + (4(k+1) - 1) = (k+1)(2(k+1) + 1) = (k+1)(2k + 3)$. By the inductive hypothesis we can see that the left-hand side is equivalent to $k(2k+1) + 4k + 3$. If we expand, we can see that $k(2k + 1) + 4k + 3 = 2k^2 + 5k + 3$, which can be factored into $(k + 1)(2k + 3)$. Notice that this equals the target right hand side, thus by the principle of mathematical induction we have proved that $3 + 7 + 11 + \dots + (4n-1) = n(2n + 1)$ for all positive integers n .